

Reanimating Images using Neural Representations of Dynamic Stimuli

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Abstract

While computer vision models have made incredible strides in static image recognition, they still do not match human performance in tasks that require the understanding of complex, dynamic motion. This is notably true for real-world scenarios where embodied agents face complex and motion-rich environments. Our approach, BrainNRDS (Neural Representations of Dynamic Stimuli), leverages state-of-the-art video diffusion models to decouple static image representation from motion generation, enabling us to utilize fMRI brain activity for a deeper understanding of human responses to dynamic visual stimuli. Conversely, we also demonstrate that information about the brain’s representation of motion can enhance the prediction of optical flow in artificial systems. Our novel approach leads to four main findings: (1) Visual motion, represented as fine-grained, object-level resolution optical flow, can be decoded from brain activity generated by participants viewing video stimuli; (2) Video encoders outperform image-based models in predicting video-driven brain activity; (3) Brain-decoded motion signals enable realistic video reanimation based only on the initial frame of the video; and (4) We extend prior work to achieve full video decoding from video-driven brain activity. BrainNRDS advances our understanding of how the brain represents spatial and temporal information in dynamic visual scenes. Our findings demonstrate the potential of combining brain imaging with video diffusion models for developing more robust and biologically-inspired computer vision systems. We show additional decoding and encoding examples on this site: <https://brain-nrds.github.io/>.

1. Introduction

Human vision excels at understanding complex, dynamic scenes, while computer vision offers powerful tools for data processing and analysis. This work explores how insights from human brain activity, specifically in response to dynamic visual stimuli, can enhance the ability of computer vision systems to understand complex motion. Critically, as inherently embodied systems, our brains have evolved

to process and represent the daily experience of naturally motion-rich environments in ways that are both semantically grounded and behaviorally-relevant. For example, a video of a person walking affords information not only about the visual features of the walker, but also about motion patterns, context, intent, and relationships between the different elements in the scene (e.g., sudden increase in gait implies urgency). This interplay between spatial and temporal information is impossible to capture from static images; accordingly, humans excel at inferring such high-level information from scene dynamics [5].

Recent advances have improved our understanding of how visual information is represented in the brain through computer vision models [6–8], yet a significant gap persists in explaining how the brain processes dynamic visual stimuli. While some studies have started to explore the neural representation of dynamic visual stimuli [3, 9, 10], there remains substantial potential to further harness modern computer vision techniques, especially video-based models. There has also been little work demonstrating how insights from human vision can be leveraged to build better computer vision systems. Here we address both challenges, demonstrating that state-of-the-art video foundation models can be used to robustly predict human brain responses to dynamic visual inputs and that brain-decoded motion signals can help to reanimate videos based only on the first frame of the video. Critically, our framework outperforms prior work for both problems, achieving lower end point error (EPE) for motion decoding based on brain responses as compared to [3] and better motion prediction based on initial static frames with brain data than without brain data.

In this paper, we present a framework for analyzing dynamic visual stimuli that explicitly separates the modeling of static and motion-related image representations as they are processed in the human brain: Neural Representations of Dynamic Stimuli (“BrainNRDS”). By decoupling these elements, BrainNRDS offers a targeted approach to understanding the distinct neural pathways used to process and represent dynamic visual information in real-world scenes. Better specified models of dynamic visual processing are also critical for developing effective models of ecologically critical domains such as social interaction [5], action rep-

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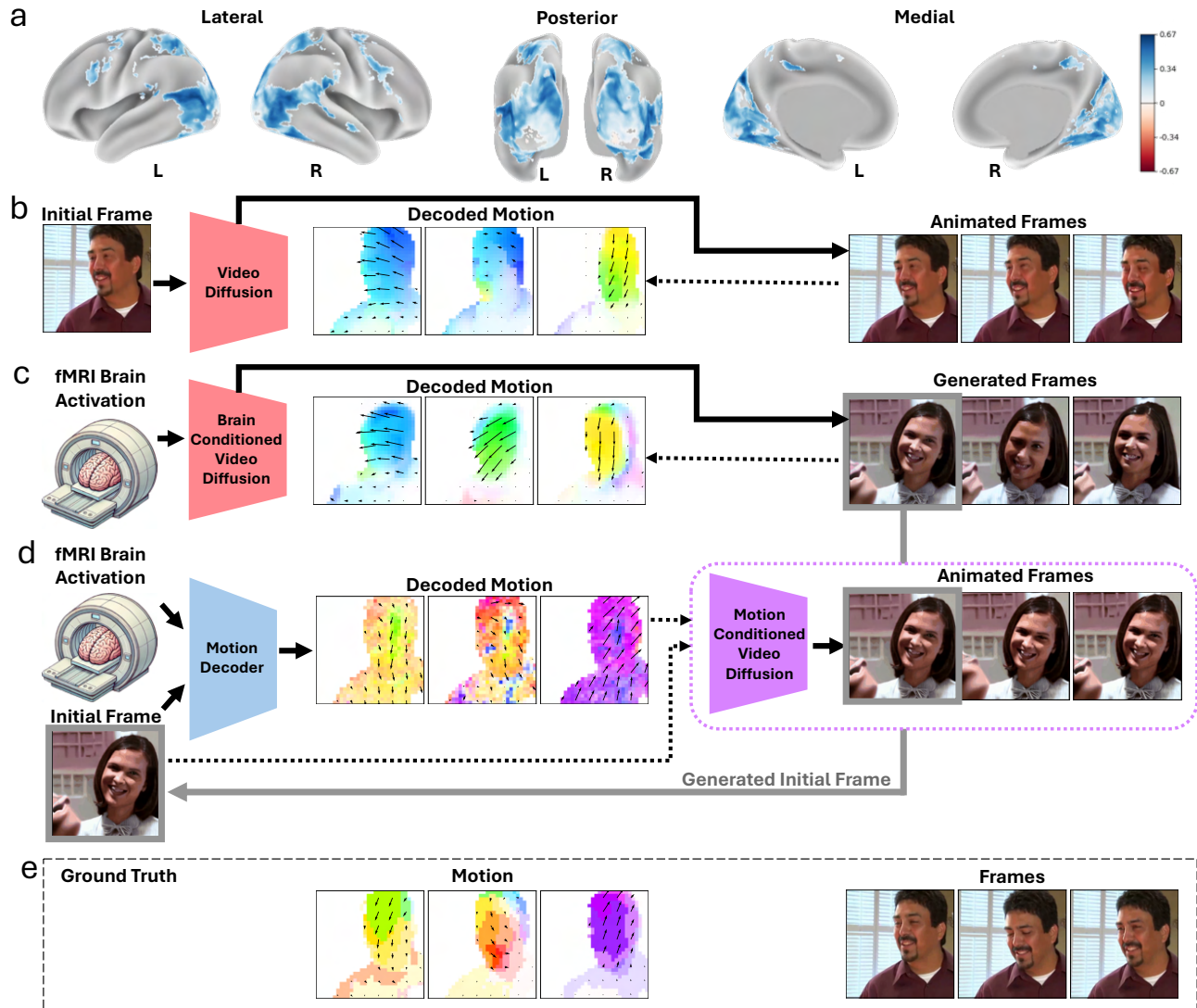


Figure 1. **Encoding and decoding video motion using brain activity.** (a) fMRI brain activity can be predicted (using a Pearson coefficient) using off-the-shelf video encoders (VideoMAE [1]) extracted from the viewed video. In the converse direction, we can generate video by decoding brain activity. (b) Many existing video diffusion models (e.g., SVD [2]) generate a video by animating an initial frame. (c) This suggests that “brain-to-video” generation can be achieved by fine-tuning diffusers to condition on fMRI input (e.g., MindVideo [3]). (d) In our approach we explicitly decouple the task of image and motion generation from brain activity. Given an initial video frame (which could be decoded from brain activity as in (c)) and fMRI input, we train a network to predict optical flow. We then animate the initial frame by feeding the predicted flow into an off-the-shelf motion-conditioned diffusion model (e.g., DragNUWA[4]). Our disentangled pipeline produces more accurate brain-conditioned motion decodings than either (b) or (c). (e) Ground truth optical flow and video.

resentation, and dynamic causal structure [11] – none of which are easily extracted from static images.

Contributions. First, we demonstrate that visual motion information in the form of optical flow can be predicted (or decoded) from brain activity as measured by fMRI. Second, we show that this predicted motion can be used to realistically animate static images using a motion-conditioned video diffusion model in which the motion is driven by brain activity. Moreover, we demonstrate that brain activity can produce more reliable animations than image-

conditioned video diffusion models trained on massively-large datasets. Third, we apply BrainNRDS to existing methods that decode the viewed video by using their initial frame to do full video decoding from video-driven brain activity and find that we more accurately predict the viewed motion than existing methods. Finally, we show prediction in the reverse direction: recent video encoders (e.g. VideoMAE and VC-1 [1, 12]) more effectively predict fMRI brain activity from videos compared to image encoders. Overall these results demonstrate the effectiveness of BrainNRDS

as a novel, extensible framework for interpreting how the human brain processes dynamic visual information.

Larger Objectives. Our project also has several larger objectives. First, developing high-performing models for motion decoding based on neural data (Figure 1d) is an important precursor to building effective encoding models [13]. Second, the finding that VideoMAE is able to accurately predict brain responses (Figure 1a) is an important step towards building more general and informative video encoding models for brain prediction [7]). Encoding models are also important components in comparisons between model-based representations and empirical neural data, which can reveal key insights into which visual features the brain prioritizes and how they align with computational processing in interpretable artificial neural networks [14]. Third, we apply BrainNRDS to existing methods which decode the viewed video to do full video decoding from video-driven brain activity [3] and find that we more accurately predict the viewed motion than existing methods. Fourth, although comparison points are limited due to the underexploration of this research area, our finding that neural data improves video prediction performance better than other extant models is a simple demonstration of the point that latent information encoded in brain data may be useful for improving the performance of computer vision systems [15]. This approach may prove to be especially powerful for domains that are inherently anchored in human behavior, for example, dynamic social interactions.

2. Related Work

Static Visual Stimuli Decoding from fMRI. Reconstructing visual images from human brain activity, such as that measured by functional magnetic resonance imaging (fMRI), has significantly advanced through the use of deep learning models trained on natural image datasets [6, 8, 16–30] and methods that leverage generative models to decode brain representations [3, 31–57]. This progress is largely due to the availability of large, publicly accessible fMRI datasets with well-annotated stimuli, such as BOLD5000 [58] and the Natural Scenes Dataset (NSD) [59], along with multimodal generative models that have been pre-trained on very large datasets, such as Stable Diffusion [60].

Dynamic Visual Stimuli Decoding from fMRI. While there has been extensive decoding work on image reconstruction from fMRI, there has been much less work on video reconstruction. Within the work on decoding motion from fMRI, much of prior work focuses on classifying global motion in artificial stimuli like random dots moving in a coherent direction [61, 62]. Recent work has also explored disentangling static and motion features using semantically meaningful dot movements [63]. We significantly advance this by decoding motion in complex, naturalistic visual stimuli at a more fine-grained, object-level

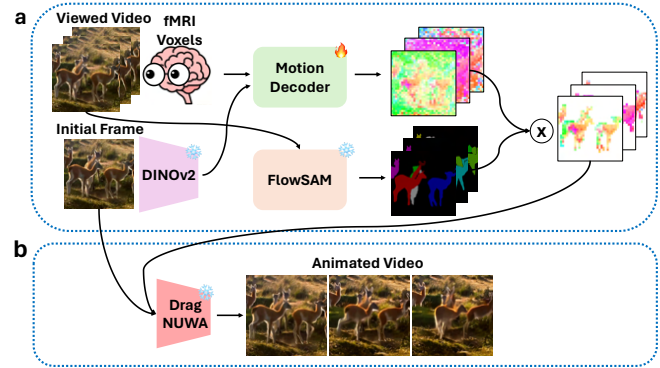


Figure 2. BrainNRDS pipeline for motion decoding and video generation. (a) BrainNRDS takes in neural data and image features from DINOv2 [68], extracted from the initial frame to predict consecutive future dense optical flow fields. Salient objects are masked out using FlowSAM [69] to obtain the masked object flow. Snowflakes=frozen; flames=actively trained. (b) The initial frame is realistically animated using our predicted motion and a motion-conditioned video diffusion model, DragNUWA [4].

resolution. This allows us to study how the brain represents motion in more ecologically valid scenarios.

Our fMRI-based approach offers a whole-brain perspective compared to single-neuron recordings, which are limited in spatial coverage [61, 62, 64–67]. Despite the challenges of fMRI’s slow temporal resolution, our results demonstrate successful decoding of motion information from complex, naturalistic stimuli at the object level. Thus, fMRI, despite its limitations, can enable valuable insights into the neural representation of dynamic information.

Concurrent to motion decoding research, there is a parallel branch of research focusing on decoding the raw video from fMRI activity [3, 70–74]. Methods like MindVideo and NeuroClips [3, 73] train to align temporal semantics from fMRI and generate frames across time. Notably, our approach explicitly disentangles the representations of motion information (represented as optical flow) from static images in fMRI signals. This allows us to differentiate our present work from previous work by enabling interpretability. Specifically, we isolate and focus on decoding motion (as optical flow) from fMRI signals, as opposed to more difficult to interpret mixture of static image features and motion. This disentanglement allows us to learn an explicit representation of motion which can be directly compared to groundtruth and be more directly related to spatially localized neural mechanisms, as opposed to an entangled mixture of static image and dynamic features. In contrast, NeuroClips [73] decodes blurred video as a proxy for motion. Comparing the two approaches, blurred video preserves rough scene composition, while optical flow explicitly encodes motion magnitude and direction, offering links to neural motion processing. At the same time, we ensure temporal consistency by integrating predicted optical

flow with a motion-conditioned video diffusion model.

3. Methods

In order to understand how the brain processes dynamic visual stimuli, we focus on three directions. In the first, we examine the motion representation contained in the brain by decoding dense optical flow conditioned on the initial frame. In the second, we visualize the predicted motion by reconstructing the observed dynamic visual stimuli by leveraging a motion conditioned image diffusion model (DragNUWA). Finally, we examine which visual features, (static vs dynamic; semantic vs visual similarity) best explain brain activity on a voxel-wise level.

3.1. Motion Prediction

We extract the information relevant to visual motion in order to predict the optical flow in videos observed by participants. To enhance interpretability, we’ve architecturally separated static image features from dynamic features in BrainNRDS. By conditioning the model on the initial frame alongside neural data, we enable our model to focus on extracting dynamic visual information. This approach improves interpretability compared to models like MindVideo, which simultaneously predict both dynamic and static features. In contrast, our approach disentangles the specific contributions of dynamic components.

At a given fMRI scan at index i we have fMRI voxels $B_i \in \mathbb{R}^n$ and corresponding video frames $\mathcal{I}_i \in \mathbb{R}^{T \times H \times W \times 3}$. We obtain optical flow fields $O_i \in \mathbb{R}^{(T-1) \times H \times W \times 2}$ from RAFT [75], which consist of $(T - 1) \times H \times W$ optical flow vectors in $\mathbb{R}^2$. Prior work [76] has found that reformulating optical flow prediction as classification improves performance. We take a similar approach and quantize the optical flow vectors into a codebook of k clusters using k-means. We obtain image features with pre-trained vision model G . Concretely, we learn a function M_θ that classifies the fMRI voxels B_i and image features of the corresponding initial frame $G(\mathcal{I}_{i,1})$ to the quantized future predicted optical flows fields: $M_\theta([B_i, \mathcal{I}_{i,1}]) \rightarrow O_i$.

In order to focus our evaluation on salient objects in a scene, we apply masking using FlowSAM [69] to track and identify salient objects.

Visualizing the Predicted Motion. We can visualize the predicted motion by passing in the predicted optical flow field and the initial frame to a pretrained motion-conditioned video diffusion model, DragNUWA [4]. This allows us to generate the dynamic vision stimuli (observed by the participants) or (driving the neural response).

3.2. Visual Stimuli-To-Brain Encoding

Here we describe a complementary approach to understanding dynamic visual stimuli processing in the brain. We consider models trained on different tasks:

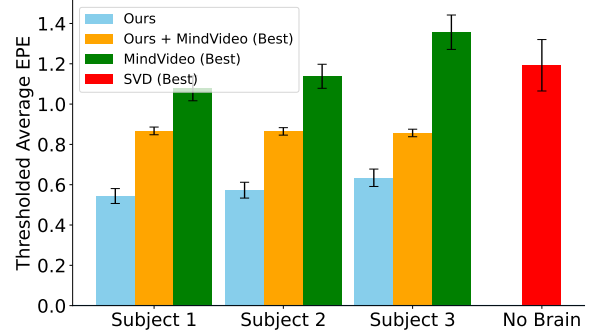


Figure 3. **Quantitative motion decoding.** Optical flow predictors trained with neural data (Ours) are statistically better than both generative models trained without neural data (No Brain - Stable Video Diffusion (Best) [2]) and generative models that fail to disentangle appearance and motion (MindVideo (Best) [3]). We find that our method conditioned on the initial frame generated by MindVideo (Ours + MindVideo (Best)) better predicts the optical flow than MindVideo. We average the end point error over the predicted and ground truth masked optical flow vectors. Similar to [77], we report end point error on pixels whose ground truth flow magnitudes exceed 1% of the pixel width of the frames. Lower values are better. Paired t -tests comparing end point error for “Ours” versus SVD (Best) for each participant are as follows: S1: $p \leq 1.197\text{e-}8$; S2: $p \leq 3.157\text{e-}8$; S3: $p \leq 9.140\text{e-}7$. Error bars represent the standard error of the mean.

Method	Video-based		Frame-based	
	VideoMAE	CosSim↑	CLIP CosSim↑	Pixel SSIM↑
MindVideo	0.742±0.006	0.879±0.004	0.171±0.02	
Ours end-to-end	0.769±0.006	0.896±0.003	0.214±0.01	
Subject 1	0.771±0.006	0.896±0.003	0.218±0.01	
Subject 2	0.774±0.006	0.896±0.003	0.2±0.01	
Subject 3	0.762±0.006	0.894±0.003	0.223±0.01	

Table 1. **Video generation quantitative results.** Our end-to-end method uses the initial frame generated from fMRI by MindVideo and generates the video solely from fMRI without access to GT frames. We report semantic fidelity metrics and SSIM on both video and frame level. Results are significant ($p < 0.05$ using two-sample t -test) compared to MindVideo.

1. **Static Image:** These models are trained on static visual stimuli without guidance from dynamics. The features extracted can identify brain voxels that are semantically and spatially tuned.

- (a) **Supervised:** ResNet-50 is trained on ImageNet to categorize images into semantic classes [78].
- (b) **Self-supervised:** DINOv1 and DINOv2 are vision transformer-based models trained to learn similar representations for visually similar patches of images without supervision on semantic similarity [68, 79].
- (c) **Semantically-supervised:** CLIP ViT-B/32 and CLIP ConvNeXt are trained to learn align semantics and images. CLIP ViT-B/32 uses a visual transformer architecture and CLIP ConvNeXt uses a convolutional backbone, sharing inductive biases with ResNet-50 but

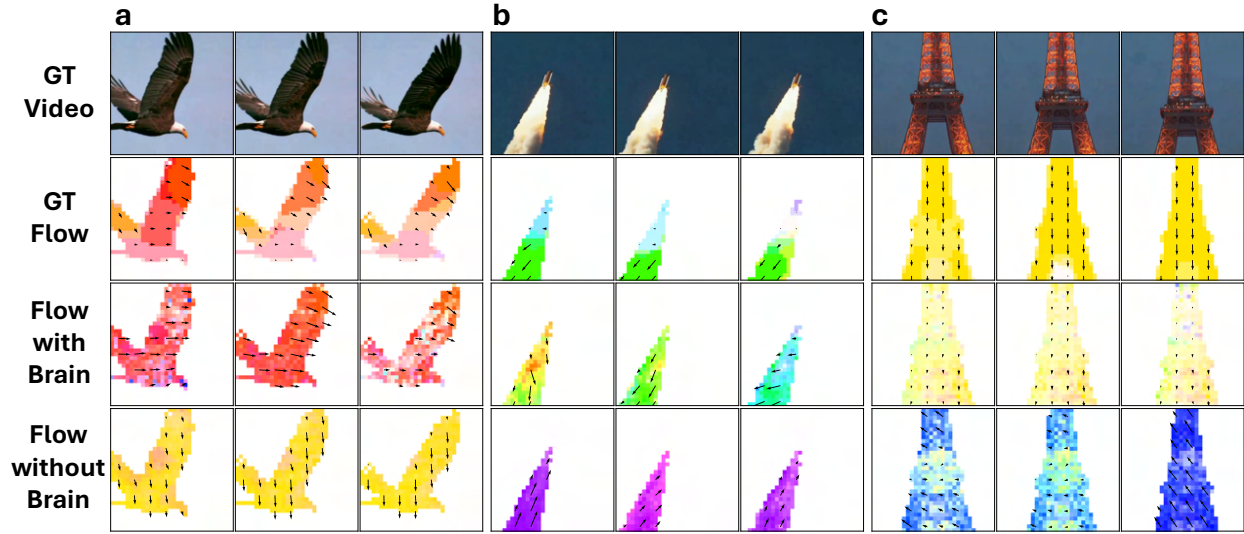


Figure 4. **Motion predicted with and without neural data.** Motion predicted using brain data and a static reference image is qualitatively both plausible and aligned with the original video compared to motion predicted from only the reference image using Stable Video Diffusion [2]. Motion predicted from purely image features is plausible but not aligned with the original video. In this latter case, the model is hallucinating flows. (a) An example video where there are multiple plausible actions. (b) An example video where camera motion overrides the plausible motion. (c) An example video where there is ambiguity of stationary objects due to camera motion.

trained with a different objective. CLIP ViT-B/32 AF and CLIP ConvNeXt AF consist of features Averaged over Frames (AF) spanning the video [80, 81].

2. **Video:** These models are trained on dynamic visual stimuli. The features extracted can identify brain voxels that are spatially and dynamically tuned.

(a) **Self-supervised:** VideoMAE, VideoMAE Large, Hiera Base Plus, and Hiera Huge are vision transformer-based architectures trained on Kinetics 400 with a masked autoencoder loss. The encoder learns embeddings used for downstream video tasks such as action recognition [1, 82].

(b) **Semantically-supervised:** X-CLIP is a CLIP model fine-tuned with cross-frame attention to learn temporal relations [83].

3. **Embodied AI:** These models are trained to align representations of single frames across time for embodied AI visuomotor manipulation. The features extracted can identify brain voxels tuned to action representation.

(a) **Self-supervised:** VIP is a ResNet-50 based architecture trained on Ego4D. VC-1 is a vision transformer-based architecture trained on seven human egocentric video datasets, including Ego4D. For both models, we also test an encoder trained on features averaged over frames spanning the video (VIP AF and VC-1 AF [12, 84]).

(b) **Semantically-supervised:** R3M is a ResNet-50 based architecture trained on Ego4D to align frames that are close in time and semantically similar [85].

We train ridge regression models on visual features from

different encoding models to predict voxel-wise responses.

4. Experiments

In 4.1, we describe the dataset and preprocessing used. In 4.2 we explore motion decoding (reconstruction) conditioned on initial image and brain information using Brain-NRDS, and augment quantitative analysis with visualizations derived from a motion-conditioned diffusion model. In 4.3, we explore the performance of different visual backbones in predicting brain activations.

4.1. Dataset

We utilize the Dynamic Natural Vision dataset [86], a public, IRB approved fMRI dataset composed of brain responses from 3 participants viewing diverse naturalistic video clips. The dataset contains 11.5 hours of fMRI data, with the 3 participants viewing the same videos. The fMRI data was recorded using a 3T MRI system with a 2 second TR and the videos were presented at 30 FPS. 18 8-minute videos were seen twice and 5 8-minute videos were seen 10 times, including the train and test sets which have no overlapping videos. We use the same train/test split described in [86]. The fMRI data is preprocessed and aligned with presented stimuli using the pipeline from the Human Connectome Project. We further normalized the fMRI activations so that each voxel’s response was zero-mean and unit variance on a session basis. Then we average the response across repeated viewings of the same visual stimuli.

In practice, we temporally downsample the flows to $c = 3$ frames, selected in equal intervals over the corre-

sponding 2 second fMRI brain scan, and aim to predict the c quantized consecutive optical flow fields. We also spatially downsample the flow to a resolution of 32×32 and quantize the flows into 40 clusters. We extract image features from the first frame using DINOv2. During test time, we take a weighted-sum over all the classes per pixel, following [76]. We train the model using cross entropy loss. We illustrate our model’s pipeline in Figure 2.

fMRI signal peaks approximately 4 seconds after stimulus presentation and slowly decays due to the hemodynamic response. In preprocessing, the fMRI data is shifted to temporally align the peak of the response to the onset of its corresponding stimulus presentation. However, fMRI scans 4 seconds(2 TRs) prior and after the current TR may share information with the current TR. Consequently, when predicting the motion in the video presented at $TR=i$, we concatenate data from TRs $[i-2, i-1, i, i+1, i+2]$. Video frames are generated conditioned on the predicted motion using DragNUWA [4].

4.2. Motion Decoding

Architecture. BrainNRDS predicts motion labels by processing fMRI and image features. fMRI data is processed through an MLP then spatially broadcasted and concatenated over the image features. The concatenated features pass through three residual 1×1 convolutional blocks with dropout, while global features are extracted using AdaptiveAvgPool2d and concatenated back into the spatial features. A final 1×1 convolution classifies each spatial patch into the motion codebook.

Quantitative Flow Evaluation. We report evaluations of the decoded optical flow vectors on the end point error on the test set. We quantitatively compare the models trained on the 3 participants against baselines of: MindVideo (Best) [3], BrainNRDS leveraging the initial frame generated from MindVideo (Ours + MindVideo (Best)), and Stable Video Diffusion (SVD) [2] that does not use neural data and only the initial frame in Figure 3. We extract the optical flows from the videos generated by MindVideo and SVD using the same RAFT model we use to extract the ground truth flow. Since MindVideo and SVD are generative models that can produce multiple plausible outputs, we generate 100 samples per video for MindVideo and 10 samples for SVD (because the latter is able to condition on the true video’s first frame) and select the sample with the lowest end point error for fair comparison. We evaluate BrainNRDS using the true initial frame from the video (Ours) as well as the first frame generated by MindVideo (Ours + MindVideo(Best)). Since the SVD baseline (SVD (Best)) is not conditioned on neural data, it learns to predict plausible flows which do not necessarily align with the true flow. Indeed, we observe that our model trained on individual participant neural data performs significantly better

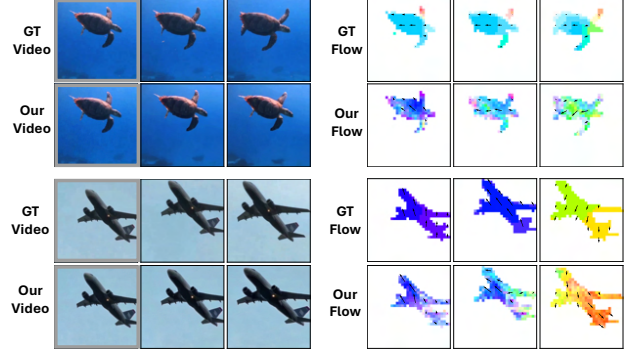


Figure 5. **Static image animation results.** Optical flow predicted from fMRI brain activity (“Our Flow”) and the ground truth initial frame (gray box) compared against the ground truth (“GT”) flow. Beneath the ground truth video frames, we show our results for animating the initial frame by combining the brain conditioned motion prediction with DragNUWA [4] (“Our Video”).

on end point error compared to SVD (Best) (end point error for individual participants – S1, S2, S3 – respectively, S1: 0.543, S2: 0.572, S3: 0.634 vs. 1.192, all $p \ll 0.001$ using paired t -tests with Holm-Bonferroni correction).

Reasoning With and Without fMRI. We present examples and explain why models trained on neural data outperform SVD (Figure 4). First, there exists ambiguity of action given static images - a frame may contain multiple possible actions. For instance, in Figure 4a we show an eagle flying. Given the first frame, the eagle could be plausibly either flapping its wings or gliding. The model trained on neural data is able to identify that the eagle is gliding. However, the model without brain conditioning incorrectly predicts that the eagle is flapping its wings.

Second, even if the action is not ambiguous, there exists camera motion in the videos which can change the direction of the flow fields. For example, in Figure 4b we show a space shuttle flying into the sky. In this case, with no camera motion the direction of motion is clearly upwards and to the right, which the model without brain conditioning correctly predicts. However, because the video is centered on the space shuttle with camera motion, the exhaust from the space shuttle appears to be going downwards and to the left. This causes the optical flow vectors to point downwards and to the left. Again, the model trained with neural data is able to correctly predict the motion of the object.

Third, there exists ambiguity of direction of motion in static images of stationary objects. For instance, if the first frame contains just a stationary portion of the Eiffel Tower, what movement should be predicted? One could choose to predict no motion, or arbitrarily predict a direction of camera motion. Indeed, we see in Figure 4c that the model without brain conditioning chooses an arbitrary camera motion pointing leftwards. However, the model trained with neural data correctly predicts the true direction which is camera

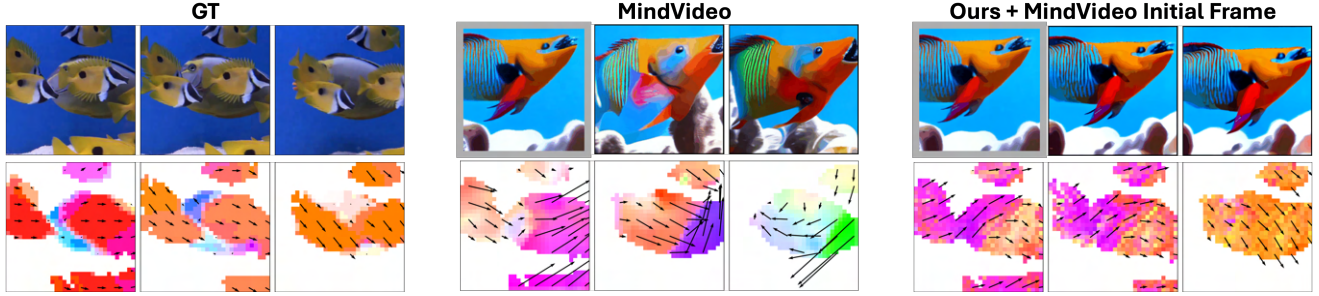


Figure 6. **Extending the method for video decoding.** Video decoding from fMRI is accomplished by leveraging the initial frame generated by MindVideo [3] box). The ground truth flow (**bottom-left**) better aligns with our predicted motion (**bottom-right**) than with the motion predicted by MindVideo (**bottom-middle**).

panning downwards. We provide additional controls and quantitative results in the supplemental.

Initial Frame Reanimation. The results derived from these scenarios indicate that the neural data contains information about the correct movement which cannot be derived from just image features.

We show motion decoding results along with the video reconstruction from DragNUWA (Figure 5). We find that the motion we predict from the fMRI brain activity is not only aligned with the ground truth motion, but also generates realistic videos. We further demonstrate that we are able to decode motion that is consistent and that suddenly changes direction, i.e. the airplane example in Figure 5.

Because we disentangle the modeling of static image and motion representations, we can extend BrainNRDS to other works that decode image frames from fMRI. We demonstrate examples of extending our method to initial frames decoded from MindVideo in Figure 6 and find we decode motion more accurately (Figure 3). We provide additional video examples in the supplemental.

Generated videos are evaluated using initial frames decoded from MindVideo (Table 1). Video-level evaluation is computed using the cosine similarity between averaged VideoMAE features of the ground truth and generated videos. Frame-level evaluation is computed using the cosine similarity between CLIP features of corresponding frames. SSIM is also reported. BrainNRDS outperforms MindVideo across all metrics, demonstrating the advantage of separately modeling spatial and temporal information.

4.3. Identifying Brain Regions Tuned to Dynamic Features

Having demonstrated the ability to predict motion from fMRI brain activity, we identified the brain regions selective for dynamic visual features. To do so, we test various vision encoding models to determine which best predict the fMRI brain activity of participants viewing dynamic visual stimuli. Specifically, we extract visual features using these models and construct voxel-wise encoding models using ridge regression [13, 87]. These voxel-wise encoding mod-

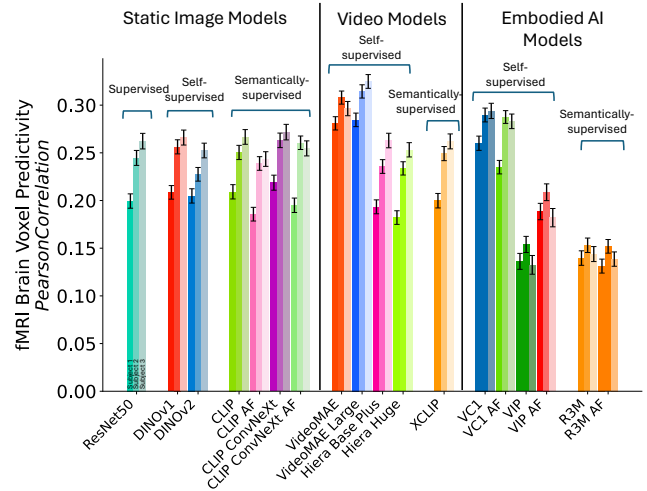


Figure 7. **Brain Encoding Prediction Performance.** We analyze the ability of visual encoders (VideoMAE [1] and CLIP ConvNeXt) to predict (encode) fMRI brain activity from visual stimuli. Self-supervised video models, VideoMAE and VideoMAE Large, best predict fMRI brain activity. Details on which brain regions are best predicted by VideoMAE Large are shown in Figure 8. For static image models, semantically-supervised models such as CLIP ConvNeXt, perform best (consistent with [7]), while VC-1 performs best among embodied AI models. Figure 9 compares the relative encoding performance of the best video (VideoMAE) and static image (CLIP ConvNeXt) models. Error bars represent the standard error of the mean.

els predict the fMRI brain activity of participants watching dynamic visual stimuli based on the extracted features.

Importantly, we test models trained under different paradigms, such as those trained on static images or video input, and models that use visual or semantic contrastive losses. By analyzing the differences in brain prediction performance, we can identify brain regions more selective to dynamic visual features.

We find that the best performing video encoding model, VideoMAE Large, outperforms the best static image encoding models at predicting fMRI brain activity (Pearson r 's for individual participants, S1: 0.285, S2: 0.314, S3: 0.324,

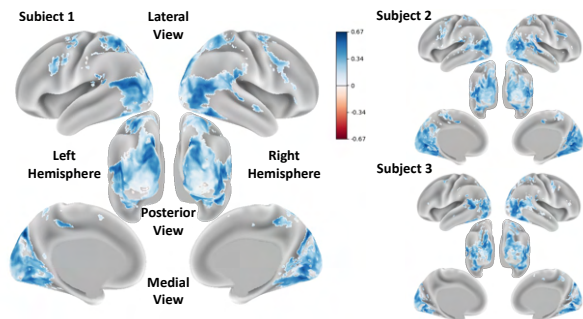


Figure 8. VideoMAE voxel-wise prediction performance. Voxel-wise fMRI voxel encoding accuracy of the best encoding model, VideoMAE Large, quantified as the Pearson correlation between measured and predicted responses. Refer to the supplemental for a labeled fMRI data parcellation of brain regions. The majority of the voxels in visual cortex are predicted well by VideoMAE. VideoMAE predicts areas TPOJ1, STV, PGi, and 7PC (all subjects, Left Hemisphere, Lateral View) particularly well – brain areas are implicated in processing visual and motor information and appear important for high-level brain functions, including motor planning and theory of mind [88]. Thus, VideoMAE may be capturing dynamic information relevant for motor planning.

all $p \ll 0.001$ by a one-sample t -test). This suggests that fMRI data during video viewing contain dynamic information which is not captured in features from static images.

Within image encoding models, CLIP ConvNeXt predicts fMRI brain activity best across all participants (Pearson r 's for individual participants, S1: 0.219, S2: 0.263, S3: 0.272, all $p \ll 0.001$), consistent with prior results [7] that alignment with language improves visual brain prediction. Averaging embeddings across frames from temporally unaligned encoding models does not enhance performance.

The best performing embodied AI vision encoding model, VC-1, outperforms the best static image encoding models at predicting brain activity (Pearson r 's for individual participants, S1: 0.260, S2: 0.290, S3: 0.294, all $p \ll 0.001$). Interestingly, on average, VC-1 outperforms other embodied AI vision encoding models across a series of tasks [12]. This suggests a possible avenue for ranking embodied AI vision encoding models on predicting fMRI brain activity. More generally, models which explicitly model temporal relationships are better at capturing neural responses to dynamic visual stimuli. We provide additional controls and quantitative results in the supplemental.

Interpreting Feature Performance. We plot voxel-wise encoding performance of the best performing model, VideoMAE [1], on an inflated cortical map to interpret the voxel-wise selectivity for dynamic visual stimuli features in Figure 8. Please refer to the caption for more detailed analysis. We identify fMRI brain voxels that are motion selective by taking the voxel-wise encoding performance difference between the best encoding video model, VideoMAE Large, and the best encoding static image model, CLIP ConvNeXt,

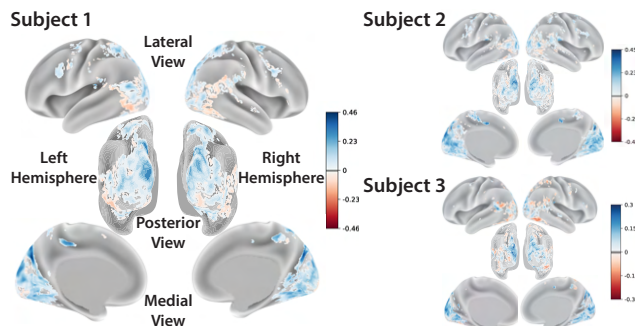


Figure 9. Voxel-wise prediction difference between VideoMAE and CLIP ConvNeXt. Voxel-wise difference of fMRI voxel encoding accuracy between the best encoding model, VideoMAE Large, and the best performing static image encoding model, CLIP ConvNeXt, quantified as the Pearson correlation between measured and predicted responses. fMRI voxels where VideoMAE predicts better are in blue and where CLIP ConvNeXt predicts better are in red. The majority of visual cortex is better predicted by VideoMAE. VideoMAE also predicts areas 5m, 5mv, 23c, 4, 24dd, and 24dv much better than CLIP ConvNeXt (S1 and S2, Left and Right Hemispheres, Medial View) – all part of the somatosensory cortex which integrates visuomotor and somatosensory information for precise motor control and spatial awareness [89, 90]. Thus, VideoMAE may be capturing action representations which are less prevalent in static image models.

illustrated in Figure 9. We find results that suggest encoding visual stimuli with VideoMAE better predicts fMRI brain activity across visual cortex and in somatosensory cortex. Voxel-wise encoding performance for each model is shown on inflated cortical maps in the supplementary information.

5. Conclusion

Limitations and Future Work. BrainNRDS successfully decodes motion of salient objects from fMRI brain activity on the Dynamic Vision Dataset [86]. However, the generalizability of our conclusions to different dataset distributions remains unclear. Future datasets with fine-grained motion can help determine the limits of motion decodability from fMRI brain activity. Furthermore, our subject-specific training approach could be improved by exploring cross-subject alignment methods, as has been shown to improve decoding performance for static stimuli [91, 92].

Conclusion and Impact. To summarize, in this paper we propose to study neural representations of motion by disentangling the modeling of static image features and motion using BrainNRDS. We show that brain activity can be used to predict visual motion, demonstrate that our predicted motion can animate static images to reconstruct the observed dynamic visual stimuli, and identify regions in the brain that are tuned for dynamic stimuli. Our results lay the groundwork for better understanding how the brain represents visual motion.

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